



CYBERSTER

NetSec Internship - Week 3 Task Report

Build and Validate a Layer 2 Lab Network

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1. 3-Switch Topology & VLAN Assignment (Deliverable 1)

A hierarchical Layer 2 topology was engineered using Cisco Packet Tracer to segment organisational traffic across distinct broadcast domains. The architecture utilises one Core/Distribution switch and two Access switches to distribute connectivity to 6 endpoint hosts.

1.1 Network Topology Design

The topology is built in two phases. Phase 1 establishes pure Layer 2 VLAN segmentation. Phase 2 introduces a Cisco 1941 Router for inter-VLAN routing using the Router-on-a-Stick (ROAS) method.

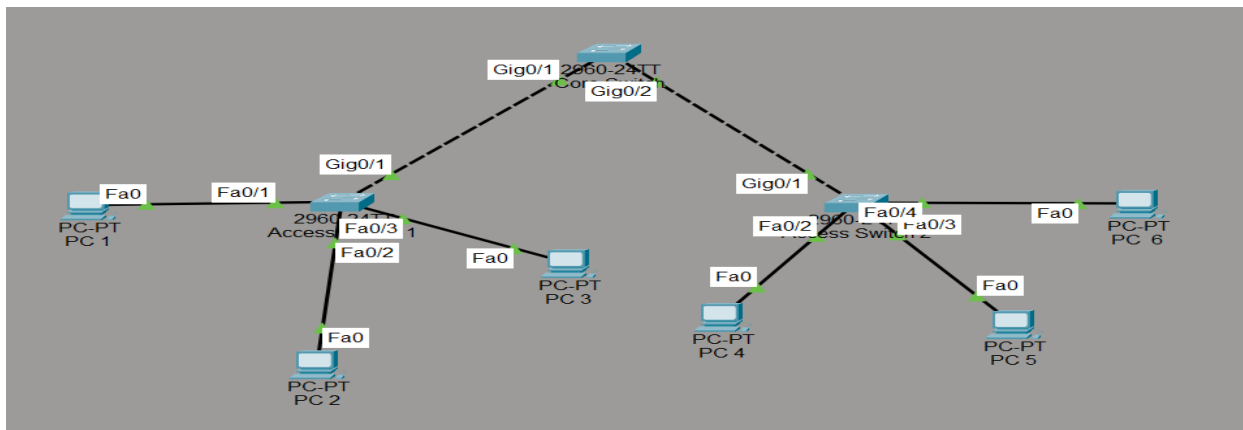


Figure 1a: Phase 1 - Layer 2 segmentation topology (no router).

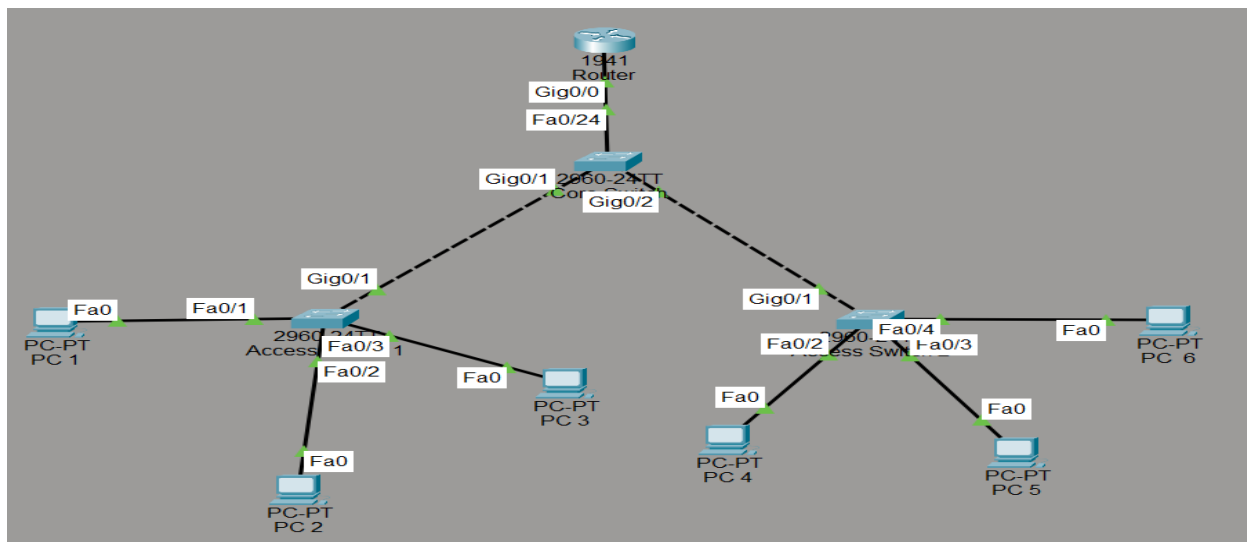


Figure 1b: Phase 2 - Router-on-a-Stick (ROAS) topology added.

1.2 VLAN Database & Subnet Architecture

Four VLANs were created to logically isolate departmental traffic. Each VLAN maps to a unique /24 subnet and is assigned a dedicated sub-interface on the router as its default gateway.

VLAN ID	Name	Subnet	Default Gateway	Hosts
10	IT_Dept	192.168.10.0 /24	192.168.10.1	PC 1, PC 5
20	HR_Dept	192.168.20.0 /24	192.168.20.1	PC 2, PC 6
30	Sales_Dept	192.168.30.0 /24	192.168.30.1	PC 3
40	Guest_Wifi	192.168.40.0 /24	192.168.40.1	PC 4

VLANs were created on the Core Switch using the following CLI commands:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT_Dept
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name HR_Dept
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name Sales_Dept
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name Guest_Wifi
Switch(config-vlan)#exit
Switch(config)#
```

Figure 2a: CLI Commands - creation of VLANs 10, 20, 30, and 40 with department names.

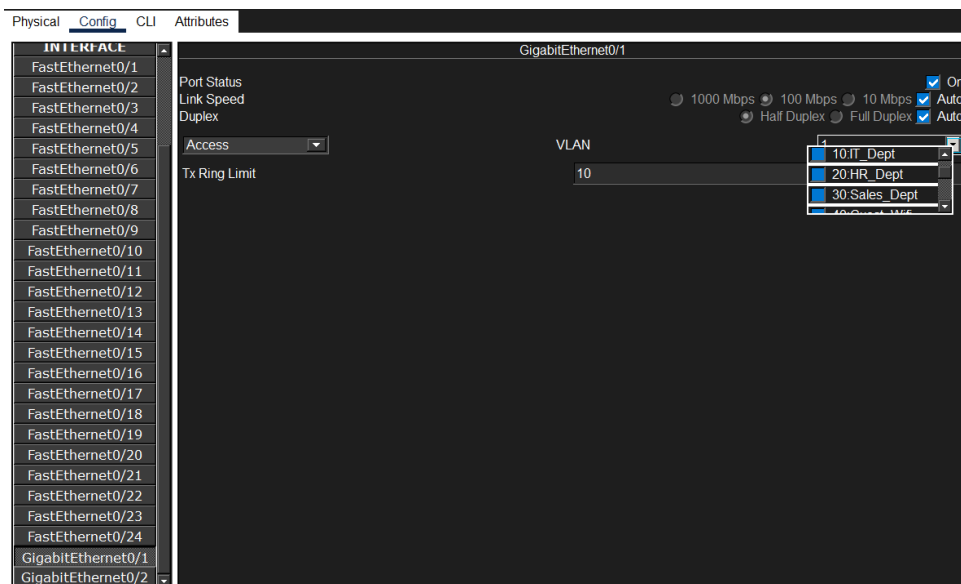


Figure 2b: Core Switch CLI - creation of VLANs 10, 20, 30, and 40 with department names.

1.3 Trunk Configuration

Trunk links carry tagged 802.1Q frames for all VLANs between the Core Switch and both Access Switches. The Core Switch uplink ports (Gig0/1 and Gig0/2) were set to trunk mode. Access Switches also required their uplink ports to be set as trunks.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range gig0/1 - 2
Switch(config-if-range)#switchport mode trunk

Switch(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
```

Figure 3a: Core Switch - Gig0/1-2 set to trunk mode (Phase 1).

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#
```

Figure 3b: Core Switch - Fa0/24 set to trunk mode (router uplink, Phase 2).

```
%LINK-3-UPDOWN: Interface FastEthernet0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk GigabitEthernet0/1 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/1 on VLAN0001. Inconsistent port type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface gig0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#
```

Figure 4: Access Switch uplink Gig0/1 set to trunk mode (PVID error resolved after trunk mode applied).

1.4 Access Port Assignment

Each PC-facing port on the Access Switches was configured as an access port and assigned to the appropriate VLAN. The table below summarises the VLAN-to-port mapping.

Device	Interface → VLAN	Department
Access Switch 1	Fa0/1 → VLAN 10	IT_Dept
Access Switch 1	Fa0/2 → VLAN 20	HR_Dept
Access Switch 1	Fa0/3 → VLAN 30	Sales_Dept
Access Switch 2	Fa0/2 → VLAN 40	Guest_Wifi
Access Switch 2	Fa0/3 → VLAN 10	IT_Dept
Access Switch 2	Fa0/4 → VLAN 20	HR_Dept

```
Switch(config-if)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#
```

Figure 5a: Access Switch 1 - port access VLAN assignments.

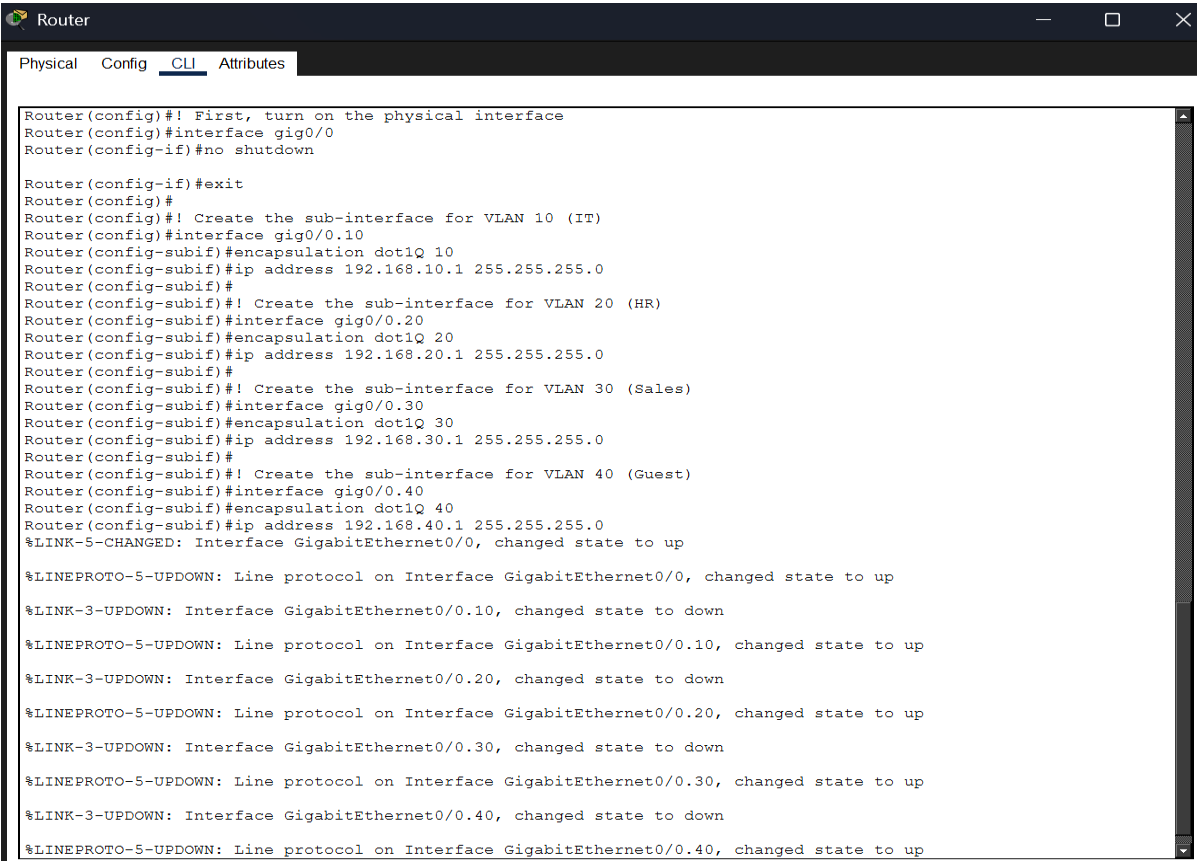
```
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#
Switch(config-if)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
Switch(config-if)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
```

Figure 5b: Access Switch 2 - port access VLAN assignments.

1.5 Router-on-a-Stick (ROAS) Configuration

To facilitate inter-VLAN routing, a Cisco 1941 Router was deployed. Its GigabitEthernet0/0 interface was divided into 802.1Q-encapsulated sub-interfaces, each serving as the default gateway for a respective VLAN. The Core Switch Fa0/24 uplink to the router was set to trunk mode.

Sub-Interface	Encapsulation	IP Address	VLAN
gig0/0.10	dot1Q 10	192.168.10.1 /24	VLAN 10 - IT_Dept
gig0/0.20	dot1Q 20	192.168.20.1 /24	VLAN 20 - HR_Dept
gig0/0.30	dot1Q 30	192.168.30.1 /24	VLAN 30 - Sales_Dept
gig0/0.40	dot1Q 40	192.168.40.1 /24	VLAN 40 - Guest_Wifi



```
Router
Physical Config CLI Attributes
Router(config)#! First, turn on the physical interface
Router(config)#interface gig0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Create the sub-interface for VLAN 10 (IT)
Router(config)#interface gig0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#! Create the sub-interface for VLAN 20 (HR)
Router(config-subif)#interface gig0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#! Create the sub-interface for VLAN 30 (Sales)
Router(config-subif)#interface gig0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#! Create the sub-interface for VLAN 40 (Guest)
Router(config-subif)#interface gig0/0.40
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.40, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up
```

Figure 6: Cisco 1941 Router CLI - sub-interface configuration for all four VLANs with 802.1Q encapsulation.

1.6 PC IP Addressing

All six PCs were statically configured with IP addresses aligned to their respective VLAN subnets. The router sub-interface IP serves as the default gateway for each host.

Host	VLAN	IPv4 Address	Subnet Mask	Default GW	Switch	Port
PC 1	10 (IT_Dept)	192.168.10.10	255.255.255.0	192.168.10.1	Access SW 1	Fa0/1
PC 2	20 (HR_Dept)	192.168.20.10	255.255.255.0	192.168.20.1	Access SW 1	Fa0/2
PC 3	30 (Sales_Dept)	192.168.30.10	255.255.255.0	192.168.30.1	Access SW 1	Fa0/3
PC 4	40 (Guest_Wifi)	192.168.40.10	255.255.255.0	192.168.40.1	Access SW 2	Fa0/2
PC 5	10 (IT_Dept)	192.168.10.11	255.255.255.0	192.168.10.1	Access SW 2	Fa0/3
PC 6	20 (HR_Dept)	192.168.20.11	255.255.255.0	192.168.20.1	Access SW 2	Fa0/4

2. Cross-Subnet ARP Packet Capture (Deliverable 2)

Layer 2 switches cannot route traffic between different subnets. Therefore, for a host in VLAN 10 to communicate with a host in VLAN 20, the originating host must first forward the packet to its Default Gateway (the Router).

To visualise this mechanism, the ARP cache on PC 1 was cleared using `arp -d`, and an ICMP echo request was initiated toward PC 2 (192.168.20.10). Because PC 1 had no cached hardware address for its gateway (192.168.10.1), it first generated a Layer 2 ARP broadcast.

Step	Event Description
Step 1 - ARP Broadcast	PC 1 sends an ARP Request (Destination: FFFF.FFFF.FFFF) to discover the MAC address of its default gateway 192.168.10.1 (the Router sub-interface).
Step 2 - ARP Reply	The Router replies with its MAC address. PC 1 stores this in its ARP cache.
Step 3 - ICMP via Router	PC 1 encapsulates the ICMP packet in a frame addressed to the Router's MAC. The frame traverses Access Switch 1 → Core Switch → Router.
Step 4 - Inter-VLAN Forwarding	The Router strips the 802.1Q tag, routes the packet to the VLAN 20 subnet, re-encapsulates it with a new VLAN 20 tag, and forwards it toward PC 2.

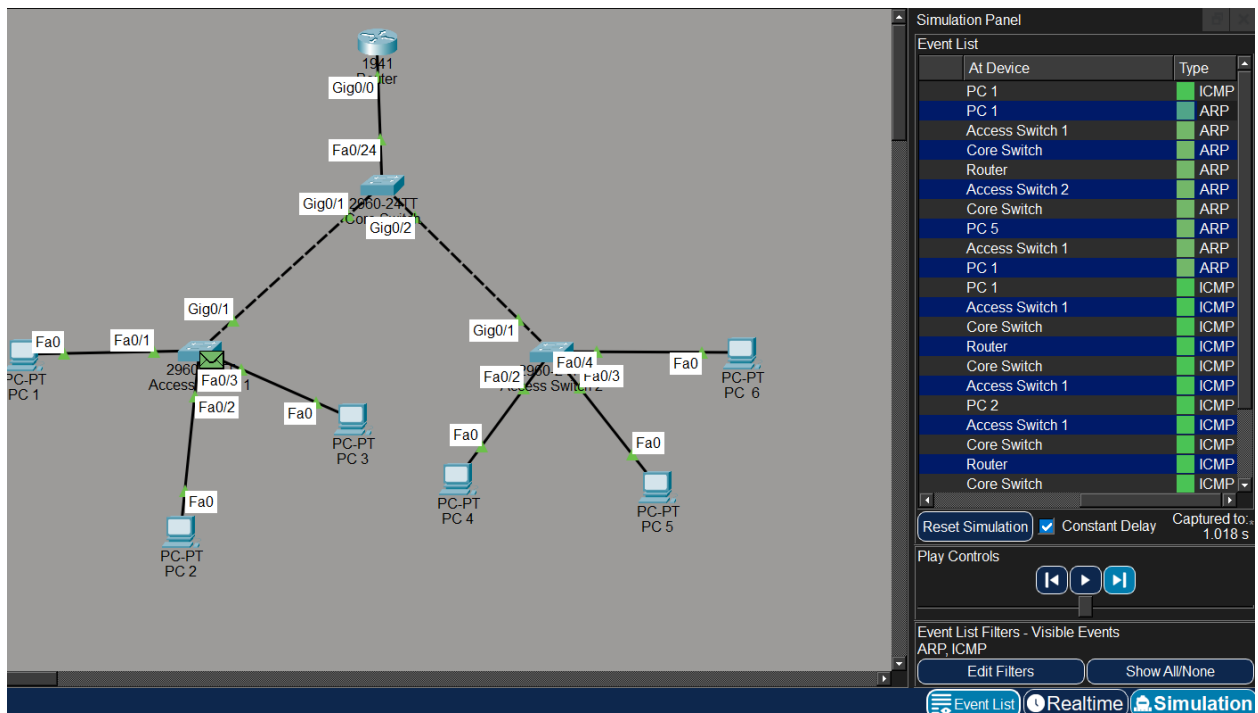


Figure 7: Packet Tracer Simulation Panel - ARP and ICMP event flow across all devices.

PDU Information at Device: PC 1

OSI Model Outbound PDU Details

Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer3	Layer3
Layer2	Layer 2: Ethernet II Header 0090.2163.7963 >> FFFF.FFFF.FFFF ARP Packet Src. IP: 192.168.10.10, Dest. IP: 192.168.10.1
Layer1	Layer 1: Port(s): FastEthernet0

1. The ARP process constructs a request for the target IP address.
2. The device encapsulates the PDU into an Ethernet frame.

Challenge Me << Previous Layer Next Layer >>

Figure 8: PDU Info at PC 1 - ARP broadcast encapsulated at Layer 2 (FastEthernet0).

3. MAC Sticky Behavior Configuration (Deliverable 3)

To harden the access layer against unauthorised endpoint connections, Port Security with the mac-address sticky feature was implemented on Access Switch 1, specifically on interface FastEthernet0/1 (connected to PC 1).

3.1 Port Security Configuration

The port was first set to access mode (port security does not operate on dynamic/auto ports). It was then configured to dynamically learn and memorise the first MAC address seen, limiting the interface to a maximum of one authorised device. The violation mode was set to Restrict to silently drop unauthorised traffic while keeping the port active.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#
Switch(config-if)#! Ensure the port is an access port (port security doesn't work on dynamic ports)
Switch(config-if)#switchport mode access
Switch(config-if)#
Switch(config-if)#! Turn port security ON
Switch(config-if)#switchport port-security
Switch(config-if)#
Switch(config-if)#! Tell the switch to dynamically learn and "stick" the MAC address to the config
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#
Switch(config-if)#! Limit the port to only 1 allowed MAC address at a time
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#
Switch(config-if)#! Set the penalty if a rogue device plugs in (restrict drops traffic but keeps the port up)
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#
Switch(config-if)#end
Switch#
```

Figure 9: CLI - access mode, MAC sticky, maximum 1 MAC, and restrict violation mode applied on Fa0/1.

3.2 Port Security Verification

Following the configuration, PC 1 generated traffic to trigger the dynamic learning process. The switch successfully bound PC 1's hardware address (0090.2163.7963) to the interface as a STATIC entry.

Parameter	Value / Description
Interface	FastEthernet 0/1 (Access Switch 1)
Port Security	Enabled
Port Status	Secure-up
Violation Mode	Restrict (drops traffic; port stays up; increments counter)
Max MAC Addresses	1
Sticky MAC	Enabled — dynamically learns and locks the first seen MAC
Learned MAC	0090.2163.7963 (PC 1)
Violation Count	0 (no violations recorded)

```
Switch#show port-security interface fa0/1
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Restrict
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 0090.2163.7963:10
Security Violation Count : 0
```

```
Switch#|
```

Figure 10: show port-security interface fa0/1 - Port Secure-up, Sticky MAC learned: 0090.2163.7963.

4. Business Risk Reduction Justification (Deliverable 4)

Implementing Port Security with the mac-address sticky feature provides critical mitigation against Layer 2 insider threats and unauthorised physical access. The table below maps specific attack vectors to the protection provided by this configuration.

Threat Vector	Attack Scenario	Port Security Mitigation
Rogue Device Insertion	Attacker unplugs a corporate PC and connects a malicious laptop to pivot into the internal network.	MAC Sticky locks the port to the first learned device. Any rogue device triggers a Restrict violation, dropping its traffic immediately.
MAC Flooding Attack (macof)	Attacker floods CAM table with spoofed MAC entries, forcing the switch into hub mode and broadcasting all traffic.	Max MAC = 1 limit prevents table exhaustion on secured ports. Even if the CAM is flooded elsewhere, port-secured interfaces remain protected.
Unauthorized Physical Access	An insider connects an unauthorized device to a switch port during a security gap.	Sticky MAC with Restrict mode silently blocks the unauthorized device without alerting the attacker, preserving forensic logs via violation counter.

Key Concept: Restrict vs Shutdown Violation Modes

Restrict: Drops traffic from unauthorised MAC. Port stays up. Violation counter increments. Traffic from authorised MAC continues uninterrupted. Preferred for production environments where stealth detection is important.

Shutdown: Immediately err-disables the entire port. Any traffic - including from authorised devices — is blocked until manual intervention (no shutdown). Used in high-security environments with zero tolerance.

5. Verification Evidence & Logs (Deliverable 5)

The logical segmentation and security configurations were validated across the Access and Core layers of the network. The following outputs confirm correct VLAN assignment, MAC learning behaviour, and end-to-end connectivity.

5.1 VLAN Database Verification

The show vlan brief command was executed on the Core Switch and both Access Switches to confirm VLANs 10, 20, 30, and 40 are active and that access ports are correctly mapped to their respective VLANs.

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
10	IT_Dept	active	
20	HR_Dept	active	
30	Sales_Dept	active	
40	Guest_Wifi	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>
```

Figure 11a: Core Switch - VLANs 10–40 active, no access ports (all trunk).

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10	IT_Dept	active	Fa0/1
20	HR_Dept	active	Fa0/2
30	Sales_Dept	active	Fa0/3
40	Guest_Wifi	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Figure 11b: Access Switch 1 - VLAN 10→Fa0/1, 20→Fa0/2, 30→Fa0/3.

```

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/2
10   IT_Dept                 active    Fa0/3
20   HR_Dept                 active    Fa0/4
30   Sales_Dept             active
40   Guest_Wifi             active    Fa0/2
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#

```

Figure 11c: Access Switch 2 - VLAN 10→Fa0/3, 20→Fa0/4, 40→Fa0/2.

5.2 CAM Table / MAC Address Verification

The show mac address-table command on Access Switch 1 confirms both dynamic and static MAC learning. The secured port Fa0/1 shows the PC 1 MAC as a STATIC entry — a direct result of the Sticky port security configuration. All other ports show DYNAMIC entries as expected.

VLAN	MAC Address	Type	Port / Note
1	00e0.a37a.1519	DYNAMIC	Gig0/1 — Core switch uplink
10	0002.1734.da01	DYNAMIC	Gig0/1 — Learned via trunk
10	0090.2163.7963	STATIC	Fa0/1 — Sticky — PC 1 (port-secured)
20	0002.1734.da01	DYNAMIC	Gig0/1 — Learned via trunk
20	0030.a378.0948	DYNAMIC	Fa0/2 — PC 2 dynamically learned

```

Switch>show mac address-table
          Mac Address Table
-----
Vlan      Mac Address      Type      Ports
----      -
          1      00e0.a37a.1519    DYNAMIC    Gig0/1
          10     0002.1734.da01    DYNAMIC    Gig0/1
          10     0090.2163.7963    STATIC     Fa0/1
          20     0002.1734.da01    DYNAMIC    Gig0/1
          20     0030.a378.0948    DYNAMIC    Fa0/2
Switch>

```

Figure 12: show mac address-table - STATIC entry on Fa0/1 (PC 1, port-secured) vs DYNAMIC entries on trunk and Fa0/2.

5.3 Connectivity & Ping Verification

End-to-end connectivity was validated using ICMP ping tests from PC 1. The intra-VLAN test (PC 1 to its own gateway) achieved 4/4 success. The inter-VLAN test (PC 1 to PC 2 across VLAN 10 → VLAN 20) achieved 3/4 success - the first packet was lost to ARP resolution, with subsequent packets routed successfully with TTL=127.

Source	Destination	Ping Result	TTL	Notes
PC 1	192.168.10.1	4/4 Success	255	Intra-VLAN: PC1 to VLAN 10 gateway
PC 1	192.168.20.10	3/4 Success	127	Inter-VLAN via Router (first pkt=ARP miss)

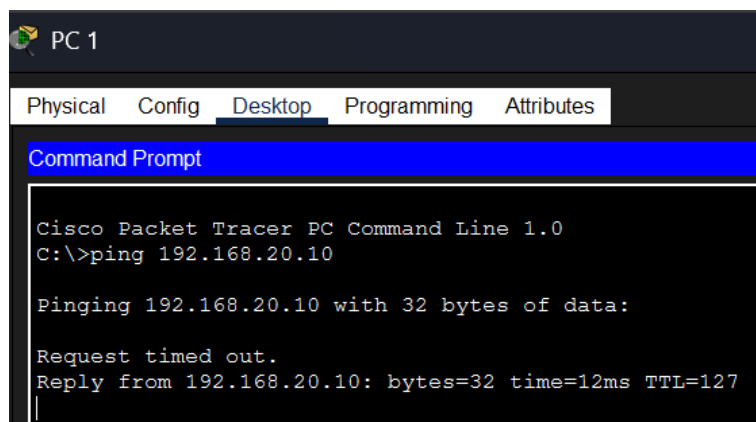
```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 13a: PC 1 → 192.168.10.1 - 4/4 packets, 0% loss (intra-VLAN gateway).



The screenshot shows the 'PC 1' configuration window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of a ping command from PC 1 to 192.168.20.10. The output indicates that the first packet timed out, but the subsequent three packets were successful, resulting in a 3/4 success rate. The TTL for the successful packets is 127.

```
PC 1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=12ms TTL=127
```

Figure 13b: PC 1 → 192.168.20.10 (PC 2) - inter-VLAN routing via Router, TTL=127.